

Meteors and the Milky Way, on a phone

Fragments of the Universe
bhapstar.com

A one-page field card. Print it, fold it, take it out with you.

GET THESE TWO RIGHT AND THE REST IS DETAIL

A small tripod

Every night mode holds the shutter open for seconds at a time. Many phones only unlock their longest exposures once the gyroscope confirms the phone is completely still, so a tripod hands you a better mode as well as a sharper frame.

A dark sky, for the Milky Way

Meteors punch through most skies. The Milky Way does not. Under city glow the background is brighter than the galaxy. An hour out of most cities is usually enough, and the change is dramatic rather than subtle.

YOUR PHONE, BY MAKE

iPhone

Lens

Main wide lens, not the ultra-wide or the zoom. Biggest sensor, fastest aperture.

Up to iPhone 16

The moon icon appears in the dark. On a tripod the exposure slider runs to about 30 seconds. Drag it to maximum and tap the shutter.

iPhone 17 series

The slider is gone. Night mode is Off, Auto or Max, from the moon icon or the camera controls button. Choose Max: up to 30s on a tripod. Auto will not give you the long exposure.

Tripod detection

Take your hands off and wait a few seconds before checking the number. Touching the phone resets it.

ProRAW

Turn it on if your model offers it, rather than a JPEG that has already decided what the sky should look like.

Samsung Galaxy

Pro mode

Manual shutter to 30 seconds, plus manual ISO and focus. Enough for both subjects.

The hidden astro mode

Download Expert RAW from the Galaxy Store. Open it, tap the settings icon, turn on Special photo options. An astrophotography icon then appears.

If you cannot find it

It only shows at 12MP resolution. Astro mode is listed for S22 and S23 series onwards, Expert RAW for S20 and later.

Star trails

Hyperlapse mode, speed 300x (marked with a star icon), then the star trails toggle appears. Set resolution to UHD. An hour of recording gives about twelve seconds of video.

Pixel and other Android

Pixel

Night Sight, on a tripod, then leave it alone. It relabels itself as astrophotography and offers up to about four minutes, stacking frames internally.

Everything else

Most Android phones have a Pro mode somewhere in the camera app. 30 seconds is plenty for both subjects.

Worth ten minutes

Several makers add their own night or light painting modes with star trails inside. Go through every mode in the camera app before deciding your phone cannot do this. The useful one is rarely on the front screen.

IF YOUR PHONE HAS A PRO MODE, SET THIS

ISO

3200

Until the sky looks grey rather than black, then back off a step

SHUTTER

20"

Long enough for a meteor, short enough to repeat often

WB

4000K

Consistent colour across every frame of the night

FOCUS

Infinity

The step most people skip. Slide it all the way by hand

FORMAT

RAW

Real latitude to lift the shadows afterwards

TIMER

3s

Stops the tap itself shaking the phone

No manual mode? You lose very little. Night mode on a tripod, fired again and again, gets you most of the way there.

LONGEST SINGLE EXPOSURE, BY ROUTE

Pixel astro mode

tripod detected, automatic

~4 min

Manual camera app

full manual, RAW, iOS or Android

60s

Samsung Expert RAW

Pro mode, manual ISO

30s

iPhone Night mode

on a tripod, no manual control

30s

Any phone, handheld

stabilisation fights you

3s

Longer is not automatically better. Four minutes gives a lovely Milky Way, though the stars will have visibly moved. For a night that includes meteors, 20 to 30 seconds repeated over and over is the sweet spot, and it still builds a fine Milky Way frame.

ON THE NIGHT

• Clean the lens

A fingerprint you would never notice by day turns every bright star into a smeared halo.

• Keep it firing

One exposure is a lottery ticket, thirty is a decent chance, a couple of hours is how people come home with the shot.

• Frame something solid

A rock, a dune ridge, a tree line, a friend looking up. Two minutes on composition pays for the whole session.

• Screen down, red light on

Eyes take about twenty minutes to adapt and a bright screen resets that in a second, for you and for anyone near you.

• Bring a power bank

The difference between stopping at midnight and staying for the best hours.

• Watch for dew

If everything suddenly looks soft, wipe the lens and carry on.

Finding the Milky Way, and framing it

The core is a summer object in the northern hemisphere. From roughly April to September it rises in the south-east after dark and arcs across the southern sky, sitting highest around the middle of the night in mid-summer. Any free stargazing app shows exactly where it sits from where you are standing, on the night you are standing there. Then give the frame something to stand on, because a foreground is what turns a record of the sky into a photograph.

The full guide, with examples

Full walkthrough with example images, plus the camera version of this guide for when you want to go further than the phone can take you.

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